**CSE 110 (Class Test 2) (Set B)**

**Total Marks: 10 Time: 15 minutes**

**Student ID:**

<table>
<tr>
<td>1.</td>
<td>What will be the output of the following code segment? (4)
<pre>
class Q1 {
   public static void main(String[] args) {
      int count = 0;
      if (count < 5) {
         System.out.println("A");
         count = count - 10;
      }
      if (count > 5) {
         System.out.println("B");
      }
      else {
         System.out.println("C");
      }
   }
}
</pre>
</td>
<td>Output:<br>A<br>C</td>
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<td>2.</td>
<td colspan="2">A number is considered 'abundant' if the sum of its factors (excluding the number itself) is greater than the number. For instance, 18 is abundant because the sum of its factors is greater than 18 (1 + 2 + 3 + 6 + 9 = 21). Write Java code to find out whether a given number (n) is 'abundant' or not. (6)
<pre>
import java.util.Scanner;
class Q2 {
   public static void main(String[] args) {
      int n;
      Scanner scanner = new Scanner(System.in);
      System.out.print("Enter n: ");
      n = scanner.nextInt();
      int i;
      int sum = 0;
      for (i = 1; i < n; i++) {
         if (n%i == 0) {
            sum = sum + i;
         }
      }
      if (sum > n) {
         System.out.println(n + " is an abundant number");
      }
      else {
         System.out.println(n + " is not an abundant number");
      }
      scanner.close();
   }
}
</pre>
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